Mohammed Altayeb Yagoub Mohammed

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Education

Ph.D., Applied Mathematics 2022–2026
Georgia State University, Atlanta, GA, USA

M.Sc., Applied Mathematics 2014–2015
African Institute for Mathematical Sciences (AIMS), Tanzania

M.Sc., Industrial and Computational Mathematics 2013–2014
University of Khartoum, Khartoum, Sudan

B.Sc., Modeling and Computational Mathematics 2004–2009
Al-Neelain University, Khartoum, Sudan

Academic Appointments

Graduate Research Assistant and Graduate Teaching Assistant 2022–2026
Department of Mathematics and Statistics, Georgia State University, Atlanta, GA, USA

- Research in mathematical epidemiology and statistical inference for dynamical systems: parameter estimation, structural and practical identifiability, uncertainty quantification, and forecast evaluation for ODE models.
- Collaborative research in computational neuroscience: conductance-based models of spinal motoneurons and neuromechanical models of quadrupedal locomotion.
- Teaching assistant for undergraduate mathematics and statistics courses (see Teaching Experience).

Lecturer of Mathematics and Statistics 2015–2022
Comboni College of Science and Technology, Khartoum, Sudan

- Taught undergraduate mathematics and statistics, including numerical analysis, linear algebra, and discrete mathematics; designed course materials emphasizing applications and computational thinking.

Lecturer of Mathematics 2015–2019
Sudan International University, Khartoum, Sudan

- Taught core mathematics courses for engineering and science programs.

Part-Time Teaching Assistant 2009–2014
Faculty of Engineering, University of Khartoum, Khartoum, Sudan

- Supported instruction in engineering mathematics, including differential equations and numerical methods.

Research Interests

Mathematical biology and epidemiology; nonlinear dynamical systems and differential equations; parameter estimation, identifiability, and uncertainty quantification; Bayesian and frequentist inference; epidemic forecasting; computational neuroscience; numerical methods; interpretable machine learning for biomedical data.

Publications

The candidate's name is shown in bold.

Peer-Reviewed Journal Articles

1. Adam A. E. Alflahi, **Mohammed A. Y. Mohammed**, and Abdallah Alsammani. A scalable transaction management framework for consistent document-oriented NoSQL databases. *Information Systems*, 140:102719, 2026. doi:10.1016/j.is.2026.102719
2. Maha H. Yousif, Abdallah Alsammani, Mohsin H. Abdalla, Eihab B. M. Bashier, **Mohammed A. Y. Mohammed**, Ali Awadallah Saeed, and Ahmed H. Fahal. Interpretable machine-learning prognosis of mycetoma from routine clinical data. *Transactions of The Royal Society of Tropical Medicine and Hygiene*, 120(9):929–947, 2026. doi:10.1093/trstmh/trag061
3. Yaroslav I. Molkov, **Mohammed A. Y. Mohammed**, Tommy Stell, Amelia Harralson, Russell Jeter, and Ilya A. Rybak. Speed-dependent turning strategies in quadrupedal locomotion: insights from computational modeling. *Frontiers in Bioengineering and Biotechnology*, 14:1787167, 2026. doi:10.3389/fbioe.2026.1787167
4. **Mohammed A. Y. Mohammed**, Hamed Karami, and Gerardo Chowell. Comparing Bayesian and frequentist inference in biological models: a comparative analysis of accuracy, uncertainty, and identifiability. *Mathematical Biosciences and Engineering*, 23(5):1156–1202, 2026. doi:10.3934/mbe.2026043
5. Yaroslav I. Molkov, Florent Krust, Russell Jeter, Tommy Stell, **Mohammed A. Y. Mohammed**, Frédéric Brocard, and Ilya A. Rybak. Ionic mechanisms underlying bistability in spinal motoneurons: insights from a computational model. *Frontiers in Cellular Neuroscience*, 19:1710893, 2025. doi:10.3389/fncel.2025.1710893

Peer-Reviewed Conference Proceedings

6. Adam A. E. Alflahi, **Mohammed A. Y. Mohammed**, and Abdallah Alsammani. Enhancement of database access performance by improving data consistency in a non-relational database system (NoSQL). In *Computational Science and Its Applications – ICCSA 2024 Workshops*, Lecture Notes in Computer Science, vol. 14816, pp. 194–205. Springer, Cham, 2024. doi:10.1007/978-3-031-65223-3_13

Preprints

7. Abdallah Alsammani, Gassan A. M. O. Farah, **Mohammed A. Y. Mohammed**, and Mehmet Yavuz. Threshold dynamics and integrated control of cholera transmission with sanitation and vaccination. *arXiv preprint* arXiv:2505.08873, 2025 (revised 2026).
8. Mozzamil Mohammed, **Mohammed A. Y. Mohammed**, Abdallah Alsammani, Mohamed Bakheet, Cang Hui, and Pietro Landi. Coexistence via trophic cascade in plant–herbivore–carnivore systems under intense predation pressure. *arXiv preprint* arXiv:2408.04862, 2024.

Manuscripts in Preparation

9. **Mohammed A. Y. Mohammed**, et al. Modeling and forecasting COVID-19 transmission across U.S. regions using epidemiological and statistical approaches.
10. **Mohammed A. Y. Mohammed**, et al. Adaptive response to disease severity: implications for vaccination and outbreak control.
11. **Mohammed A. Y. Mohammed**, Alsadig Ali, and Abdallah Alsammani. Numerical simulation and noise sensitivity analysis of plug flow reactor models: a comprehensive finite difference study.

Teaching Experience

Graduate Teaching Assistant 2022–2026
Department of Mathematics and Statistics, Georgia State University, Atlanta, GA, USA
Elementary Statistics; College Algebra; Calculus I.

Lecturer 2015–2022
Comboni College of Science and Technology, Khartoum, Sudan
Discrete Mathematics; Linear Algebra; Numerical Analysis; Algebra; Geometry; Calculus; Statistics.

Lecturer 2015–2019
Sudan International University, Khartoum, Sudan
Calculus I and II; Statistics; Differential Equations; Complex Variables; Engineering Mathematics; Numerical Analysis.

Teaching Assistant (Tutor) 2009–2014
Faculty of Engineering, University of Khartoum, Khartoum, Sudan
Differential Equations; Linear Algebra; Statistics; Vector Analysis; Numerical Analysis; Calculus.

Honors and Awards

Ph.D. Scholarship in Mathematical Sciences 2022–2026
Georgia State University, Atlanta, GA, USA

Excellence in Teaching Award 2016–2017
Comboni College of Science and Technology, Khartoum, Sudan

AIMS Master's Scholarship 2014–2015
African Institute for Mathematical Sciences, Tanzania

Presentations

- Poster presentation, Brains and Behavior (B&B) Annual Retreat, Georgia State University, Atlanta, GA, 2025.

Conferences and Workshops Attended

- Epidemic Intelligence Service (EIS) Conference, Atlanta, GA, April 22–25, 2025.
- GSU Workshop on Matrix Theory and Applications, Georgia State University, Atlanta, GA, 2023.
- Conference on Mathematical Modelling and Numerical Simulation, Al-Neelain University, Khartoum, Sudan, 2019.
- Hands-On Research in Complex Systems, International Centre for Theoretical Physics (ICTP), Trieste, Italy.

Professional Service

- Volunteer, 9th Workshop on Biostatistics and Bioinformatics, Georgia State University, Atlanta, GA, 2024.

Computational Skills

Programming:	Python, MATLAB, R, C++
Modeling and numerics:	ODE models (e.g., SEIR-type, Lotka–Volterra), numerical simulation, optimization
Statistics and ML:	Bayesian inference (Stan), maximum likelihood and least-squares estimation, time-series forecasting, scikit-learn
Tools:	Git/GitHub, LaTeX, Jupyter, shell scripting; Linux (Ubuntu) and Windows

References

Dr. Yaroslav I. Molkov
Georgia State University, Atlanta, GA, USA
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Dr. Abdallah Alsammani
Assistant Professor of Mathematics and Data Science
Delaware State University, Dover, DE, USA
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Dr. Gerardo Chowell
Georgia State University, Atlanta, GA, USA
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Dr. Alsadig Ali
Hampton University, Hampton, VA, USA
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